

Anionic Metal–Organic Framework as an Ultrafast Single-Ion Conductor for Exceptional Performance Rechargeable Zinc Batteries

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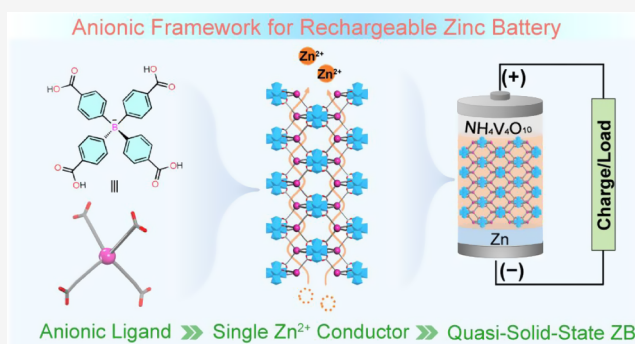


Article Recommendations



Supporting Information

ABSTRACT: Anionic metal–organic frameworks (MOFs) are promising single-ion conductive electrolytes in quasi-solid-state batteries. Herein, we propose an anionic MOF as a promising single Zn^{2+} conductor for high-performance quasi-solid-state zinc-ion battery (ZIB). The anionic MOF is synthesized by reacting carboxylic acid functionalized tetraphenylborate with a Zr_6 -oxo cluster, which displays outstanding ionic conductivity (0.83 mS cm^{-1} at -40°C , 2.75 mS cm^{-1} at 25°C , and 7.9 mS cm^{-1} at 120°C), an impressive Zn^{2+} transference number ($t_{\text{Zn}^{2+}} = 0.90$), low activation energy (0.19 eV), good interfacial compatibility, and excellent rate performance. The fabricated quasi-solid-state ZIB using $\text{NH}_4\text{V}_4\text{O}_{10}$ cathode delivers a remarkable capacity of 497 mAh g^{-1} at 0.3 A g^{-1} under ambient conditions and can maintain capacities of 420, 350, and 270 mAh g^{-1} at elevated current densities of 1, 5, and 10 A g^{-1} with both capacities and Coulombic efficiency retention at 100% after 2500 cycles. Even at a low temperature of -40°C or with a high current density of 100 A g^{-1} , excellent reversible capacities and rate performance can still be achieved. This work set up a new direction for designing and developing anionic porous materials as advanced solid electrolytes for zinc and/or other secondary batteries.



INTRODUCTION

The ultimate goal of electrochemical energy storage research is to develop environmentally friendly, high-capacity, and safe rechargeable batteries.^{1–3} Among the various battery systems, rechargeable zinc-ion batteries (ZIBs) have emerged as a promising candidate for the postlithium era by virtue of their low electrode potential (-0.76 V vs SHE), high theoretical capacity (820 mAh g^{-1}),^{2,4–6} abundant natural reserves, and eco-friendliness.¹ Despite these merits, their practical application is persistently beset by many challenges, such as zinc corrosion, dendrite formation, hydrogen evolution reaction (HER) in aqueous systems, and freezing tendency of aqueous electrolytes at low temperatures.^{7–9} Electrolyte is the main component of ZIB,¹⁰ and the development of solid-state electrolytes (SSEs),^{11,12} which have been extensively investigated in lithium batteries for their mechanical barriers,¹³ has been proposed to alleviate or overcome these problems.¹⁴ However, their low Zn^{2+} ion conductivity, poor interface compatibility with electrodes, and insufficient chemical/electrochemical stabilities still pose a formidable challenge for commercial applications.^{15,16}

Metal–organic frameworks (MOFs) have recently emerged as promising SSEs for rechargeable batteries due to their designable and regular porous structures,^{17,18} which facilitate the incorporation of high-density charged species and provide additional space for guest molecules as the electrolyte within a confined geometric volume.^{19–21} However, nearly all MOF-

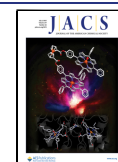
based SSEs need to incorporate extrinsic metal salts (e.g., ZnSO_4 , LiTFSI, et al.) into the frameworks,^{22–27} and the mobile anions in the pore tend to cause low transference number and undesirable polarization effect.^{28–30} Beyond that, these anions can trigger the nonuniform ion plating/stripping on the anodic metal surface, giving rise to uncontrolled mossy and dendritic metal growth.^{31,32} To avoid this, a handful of MOF-based SSEs employ postsynthetic modification to immobilize anions within the framework. Nevertheless, this strategy is restricted to MOFs with either unsaturated metal coordination sites or modifiable organic linkers, where anions can be coordinately bonded to metal clusters or covalently anchored to ligands.¹⁷ Although promising, this necessarily means that not all metal clusters or ligands can be modified to provide continuous hopping sites for charge carriers,³⁶ and even worse, the requirement of the extrinsic metal salts is sometimes even more than the amount of available binding sites.^{34,37} Consequently, these MOF-based SSEs still suffer from relatively low ionic conductivity and transference

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number, leading to the fabricated batteries still exhibiting poor rate performance and high resistance.^{32,34,35,38}

The synthesis and application of anionic MOFs as single-ion conductors represent a promising strategy to overcome these inherent limitations.^{39–43} We previously developed an anionic tetraphenylborate (BPh_4^-) ligand and employed it to construct an anionic MOF as a single Li^+ conductor for quasi-solid-state lithium battery.⁴⁴ We herein demonstrate the successful extension of this strategy by using anionic MOF as an efficient single-ion conductor for rechargeable ZIB. We first successfully targeted a robust Zr-based anionic framework using the carboxylic acid functionalized lithium tetraphenylborate anionic ligand. Through ion exchange of its intrinsic Li^+ ion with $\text{Zn}(\text{H}_2\text{O})_6^{2+}$, the resulting single-ion Zn^{2+} conductor exhibited excellent ionic conductivity, impressive Zn^{2+} transference number, and low activation energy. This remarkable efficiency could be maintained across a wide temperature range from -40 to 120 $^\circ\text{C}$, with ionic conductivities all satisfying the stringent requirements of the solid-state ZIBs. At room temperature, the anionic MOF-based $\text{NH}_4\text{V}_4\text{O}_{10}$ solid-state ZIB delivered a remarkable capacity of 497 mAh g^{-1} at 0.3 A g^{-1} and could maintain the capacities of 420 , 350 , and 270 mAh g^{-1} when the current density increased to 1 , 5 , and 10 A g^{-1} with both capacities and Coulombic efficiency retention at 100% after 2500 charge–discharge cycles. Notably, even under some extreme conditions, such as low temperature (-40 $^\circ\text{C}$) or high current density (100 A g^{-1}), our solid-state ZIB could still display excellent rate performances. Lastly, our anionic MOF electrolyte also exhibited good compatibility with various cathodes like MnO_2 , $\text{Cu}_3(\text{HHTP})_2$, and PTCDA (3,4,9,10-perylenetetracarboxylic dianhydride), enabling all corresponding ZIBs to display superior rate performance. This work establishes anionic MOFs and other anionic porous materials as a versatile platform for the development of next-generation solid-state batteries.

RESULTS AND DISCUSSION

Anionic MOF **1** was synthesized through a solvothermal reaction between anionic tetraphenylborate ligand $\text{Li}[\text{H}_4\text{L}]$ (0.5 mmol) and $\text{ZrOCl}_2 \cdot 8\text{H}_2\text{O}$ (0.5 mmol) in a mixture of DMF (60 mL) and TFA (10 mL) at 100 $^\circ\text{C}$ for 24 h . Colorless single crystals of **1** were isolated in a 57% yield with a formula of $\text{Li}_2[\text{Zr}_6\text{O}_4(\text{OH})_8(\text{H}_2\text{O})_4\text{L}_2] \cdot n\text{G}$ based on microstructure analyses, including single-crystal X-ray diffraction (SC-XRD), Fourier transform infrared (FT-IR) spectra, elemental analysis (EA), inductively coupled plasma mass spectrometry (ICP-MS), and thermogravimetric analysis (TGA). SC-XRD analysis revealed that **1** crystallized in the tetragonal $I4/m$ space group with one-fourth of a tetraphenylborate ligand and two crystallographically independent Zr atoms in the asymmetric unit. In **1**, six Zr atoms were bonded together by four $\mu_3\text{-O}$ and four $\mu_3\text{-OH}$ groups to form the classical $\text{Zr}_6(\mu_3\text{-O})_4(\mu_3\text{-OH})_4$ cluster with four $-\text{OH}$ and four H_2O groups attached as terminal groups on the equatorial plane (Figure S1). For the T_d symmetric BPh_4^- ligand, the central B adopted sp^3 hybridization with a C–B–C angle of about 109.2 ± 0.5 and a C–B bond length of about $1.65 \text{ \AA}$ (Figure 1a). Notably, the C–B bond was slightly longer than the C–C bond ($1.56 \text{ \AA}$) in its neutral isostructure MOF-841, which was derived from the neutral ligand tetrakis(4-carboxylic acid phenyl) methane reported by Yaghi and co-workers.⁴⁵ Each BPh_4^- ligand was coordinated to four Zr_6 clusters (Figure 1b), and each Zr_6 cluster was bonded to eight tetraphenylborate ligands, thus

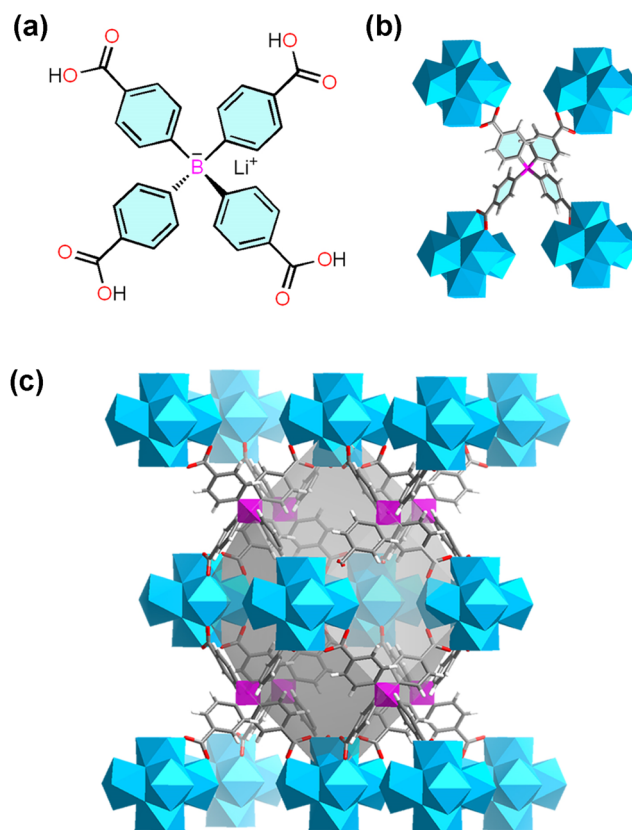


Figure 1. (a) Molecular structures of the $\text{Li}[\text{H}_4\text{L}]$ ligand. (b) Linker conformation in **1**. (c) 3D porous structure of **1**.

forming a three-dimensional (4,8)-connected anionic framework with **flu** topology. Notably, such topology was concomitant with a large dodecahedral cage with a dimension of $\sim 14.0 \times 18.2 \times 20.4 \text{ \AA}^3$ that occupied 63.7% void space of the framework for the guest inclusion (Figure 1c).⁴⁶ Unfortunately, SC-XRD could not resolve the presence of Li^+ in the framework. However, ^7Li MAS NMR analysis indicated the presence of Li^+ ; a single broad peak was observed at -1.01 ppm (Figure S4), implying the freedom of the Li^+ species in the pore that could be exchanged by other ions like Zn^{2+} to endow the framework as the solid electrolyte in the zinc solid-state battery. ICP-MS analysis of the digested sample revealed the content of Li^+ to be $2.90 \text{ wt } \%$.

The phase purity was established by the similarity between the experimental and calculated powder X-ray diffraction (PXRD) patterns (Figure S6). N_2 adsorption isotherm showed Type I adsorption behavior with the Brunauer–Emmett–Teller (BET) area of $995 \text{ m}^2 \text{ g}^{-1}$ (Figure S9). Pore size distribution derived from the nonlocal density functional theory model (NLDFT) indicated the presence of regular $\sim 8.6 \text{ \AA}$ (Figure S10), which is in good agreement with the simulation of $7.0 \text{ \AA}$ (Figure S13), thereby facilitating the rapid transport of Zn^{2+} and other ions within the pore. TGA showed that the included guest molecules in the framework could be readily removed (Figure S14). Variable temperature PXRD (VT-XRD) indicated that the structure and crystallinity integrity could be preserved up to 300 $^\circ\text{C}$ in the open air. Chemical stability was also evaluated by PXRD, N_2 sorption, and ^{11}B NMR analysis on **1** soaking in 3 M HCl (aq), 1 M NaOH (aq), and $0.001 \text{ M K}_3\text{PO}_4$ (aq) at room temperature for 24 h . These recoveries exhibited little weight loss after being

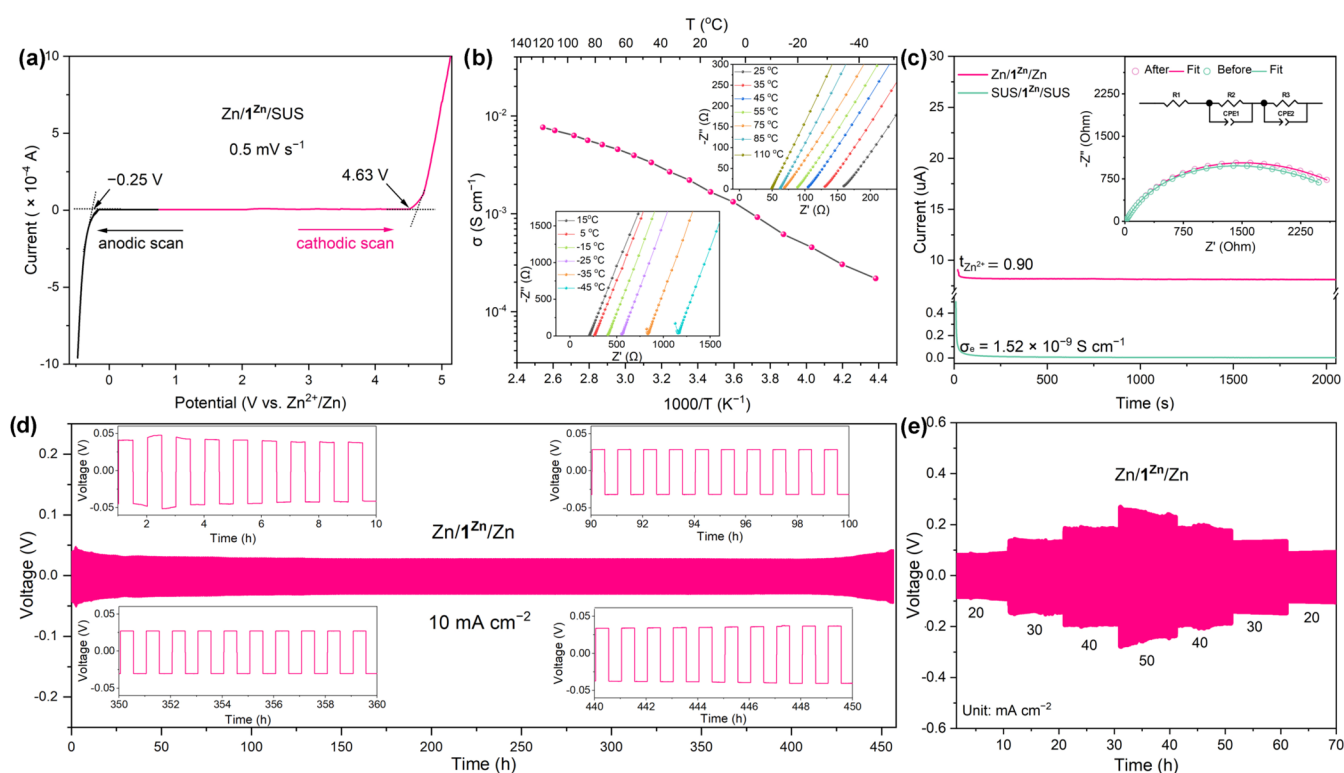


Figure 2. (a) LSV measurements for the SS/ 1^{Zn} /Zn asymmetric cells at room temperature at a scan speed of 0.5 mV s^{-1} , and IR correction was applied during LSV measurements. (b) Ionic conductivity of 1^{Zn} as a function in the temperature range from -40 to $120 \text{ }^{\circ}\text{C}$. Inset: Nyquist plots for 1^{Zn} . (c) CA curves of the Zn/ 1^{Zn} /Zn and SUS/ 1^{Zn} /SUS cells at a polarized voltage of 5 mV . Inset: Nyquist plots were obtained before and after polarization. (d) GC performance of the Zn/ 1^{Zn} /Zn symmetrical cell at a current density of 10 mA cm^{-2} at room temperature. (e) Stepped current GC curve of the Zn/ 1^{Zn} /Zn cell at room temperature.

thoroughly washed and dried. PXRD patterns were almost the same as those of the original sample after chemical stability testing (Figure S6). N_2 adsorption measurements provided BET surface areas of 762 , 697 , and $742 \text{ m}^2 \text{ g}^{-1}$, respectively, confirming the framework stability and permanent porosity (Figure S9). ^{11}B NMR analysis of these solutions after removing the anionic framework with no detectable B signal indicated neither stoichiometric nor substoichiometric amounts of decomposition during these treatments. These recoveries were then decomposed by 3 M NaOH , and the resulting ^{11}B NMR spectra were consistent with those of the $\text{Li} \cdot [\text{H}_4\text{L}]$ ligand (Figure S15), further demonstrating the full integrity of the ligands during these treatments. The electrochemical stability was then evaluated by cyclic voltammetry (CV) and linear sweep voltammetry (LSV) measurements at room temperature. CV measurement displayed a wide voltage window up to 5.0 V (Figure S18). LSV measurement exhibited a stable electrochemical window of -0.25 – 4.63 V (Figure 2a). Both CV and LSV measurements demonstrated exceptional electrochemical stability toward zinc metal and high-voltage cathode. Notably, although our framework exhibits instability toward phosphate ions with the concentration higher than 1 mM , this is not really an issue with regard to the applications in the following zinc batteries.

Considering the existence of free Li^+ ions and the readily accessible channels, we conducted ion exchange with Zn^{2+} to access the single Zn^{2+} conductor. Typically, the ion exchange was performed by immersing **1** in a 2.0 M aqueous solution of $\text{Zn}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$ at $80 \text{ }^{\circ}\text{C}$, with the solution refreshed every 4 h for a total of five cycles. After ion exchange completion, the

precipitates were filtered, thoroughly washed with water, and dried at $120 \text{ }^{\circ}\text{C}$ under vacuum to afford 1^{Zn} . PXRD showed that 1^{Zn} displayed a pattern similar to that of its pristine counterpart **1** (Figure S8). N_2 adsorption experiment provided a BET surface area of $667 \text{ m}^2 \text{ g}^{-1}$ (Figure S11). Scanning electron microscopy-energy dispersive spectroscopy (SEM-EDS) mapping analysis of a single crystal at the microstructural level indicated a uniform distribution of Zr and Zn throughout the hole crystal (Figure S17). ICP-MS measurement indicated $3.2 \text{ wt } \%$ Zn^{2+} ions incorporation with less than $0.05 \text{ wt } \%$ residual Li^+ ions. Significantly, the content of Zn^{2+} ions would not decrease even after extensive washing with water at either room temperature or $80 \text{ }^{\circ}\text{C}$, thus ruling out zinc nitrate as guest dispersing in the framework. Significantly, other cations, including Na^+ , K^+ , Ca^{2+} , Mg^{2+} , and Al^{3+} , could also be similarly incorporated into **1** through ion exchange (Figures S65–S70).

We subsequently employed 1^{Zn} as a competitive candidate for single a Zn-ion quasi-solid-state conductor, where Zn^{2+} ions migration involves intrinsic ion diffusion along anionic channels and extrinsic transport across interparticle phases. To enhance Zn-ion transportation, we trapped water into the anionic frameworks not only to hydrate the free Zn^{2+} ions but also to moisten the pores and wet the interfaces between MOF particles. After soaking 1^{Zn} in water and drying at $65 \text{ }^{\circ}\text{C}$, we obtained free-flowing 1^{Zn} powder with about $12.8 \text{ wt } \%$ water content in the anionic framework. Additionally, 1^{Zn} exhibited excellent water retention, losing less than $3 \text{ wt } \%$ after 1 week under ambient conditions and retaining $85 \text{ wt } \%$ water after 1 month. The free-flowing 1^{Zn} powder was then pressed into a densely packed morphology and sandwiched between two

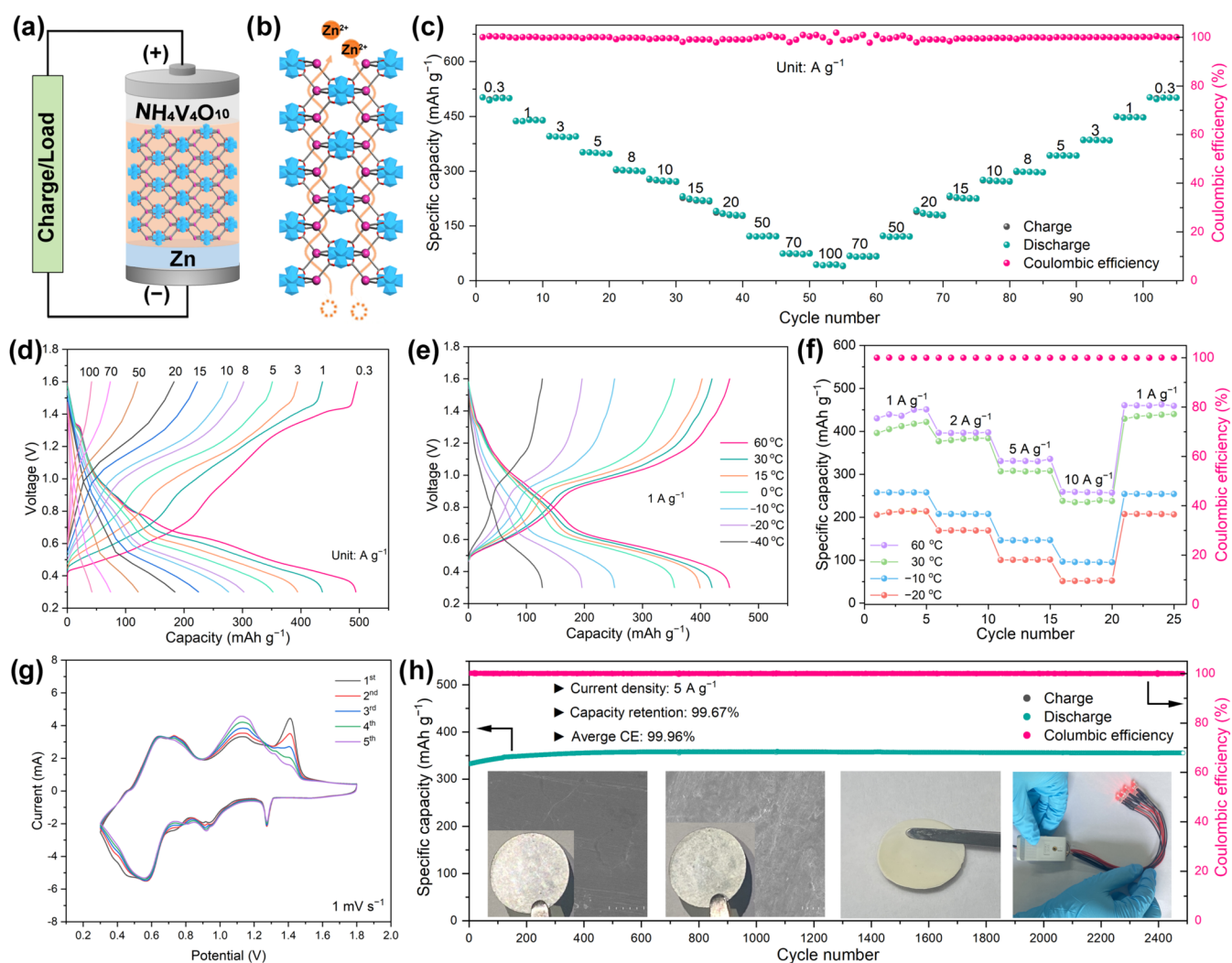


Figure 3. (a) Schematic illustration of the Zn/1^{Zn}/NH₄V₄O₁₀ SSB. (b) Schematic illustration of the single Zn²⁺ ions transportation in 1^{Zn}. (c, d) Rate and cycling performance at the current density from 0.3 to 100 A/g at room temperature. (e, f) Rate and cycling performance at the current density from 1 to 10 A g⁻¹ at 60, 30, -10, and -20 °C, respectively. (g) CV curves of the Zn/1^{Zn}/NH₄V₄O₁₀ SSB under 1 mV s⁻¹. (h) Long-term GCD performance of the battery at 5 A g⁻¹; insets: photographs and SEM morphologies of the Zn metal surface before and after the Zn/1^{Zn}/NH₄V₄O₁₀ cell cycling; the photograph of 1^{Zn} electrolyte; the obtained Zn/1^{Zn}/NH₄V₄O₁₀ SSB for lighting LED lamps.

stainless steel (SS) electrodes in an electrochemical cell for electrochemical evaluation. Ionic conductivities were assessed through alternating current (AC) impedance measurements over a frequency range of 500 kHz to 500 Hz and temperatures from -40 to 120 °C. Figure 2b displays the conductivity variation with temperature and unveils a remarkable conductivity of 2.2 mS cm⁻¹ at 25 °C, with ion conductivity increasing to 7.7 mS cm⁻¹ as the temperature rose to 120 °C, and remaining as high as 0.27 mS cm⁻¹ even when the temperature was decreased to -40 °C. We further acquired the activation energy (E_a) within this temperature range, and the value was estimated to be 0.19 eV for the fast Zn²⁺ ion transportation (Figure S22). Moreover, when 1^{Zn} was placed at room temperature for more than one year, its ionic conductivity exhibited a negligible degradation, and therefore, the ionic conductivity of the aged sample is also acceptable for practical application as the quasi-solid-state electrolyte in a zinc-ion battery.

Electronic conduction was then studied by direct current (DC) polarization measurement, yielding a value of 1.52×10^{-9} S cm⁻¹ (Figures 2c and S19). This value is 6 orders of

magnitude lower than its ionic conductivity, indicating a significant electrical insulation of the anionic framework to prevent any current flow through the electrolyte. The zinc-ion transfer number ($t_{\text{Zn}^{2+}}$) was then measured by using the Evans method, yielding a value of 0.90 (Figure 2c). The transference number of electronic conductivity was also measured with a value less than 0.01% (Figure 2c), further confirming the superior ionic conduction characteristics of 1^{Zn}. Moreover, we employed its neutral isostructure MOF-841 and the Zn-[Me₄L]₂ ester as controls to elucidate the role of Zn²⁺ ions and the anionic framework.⁴⁵ Notably, these controls showed significantly low conductivities upon hydration (2.28×10^{-7} and 1.02×10^{-8} S cm⁻¹ for MOF-841 and Zn-[Me₄L]₂, respectively, Figures S20 and S21), conclusively verifying that the observed ionic conduction was predominantly derived from the Zn²⁺ migration within the well-defined anionic framework.

Notably, the room-temperature conductivity of 1^{Zn} ranks among the highest values of all reported crystalline porous materials (e.g., COFs and MOFs).^{13,33,39,41,44} We believe this remarkable performance is due to the following synergistic

factors. First, the anionic and porous feature of **1**, derived from the BPh₄[−] linkage, establishes efficient ion transport pathways that enable rapid Zn²⁺ migration through hopping between boron centers (Figure 3b), which leads to improved ion conductivity, low activation energy, and excellent transference number. Second, the shielding effect of the coordinated water shell, in which the Zn(H₂O)₆²⁺ ions exhibit negligible bonding interactions with the anionic frameworks, further contributes to the high conductivity and low activation energy. Third, we propose that when the MOF nanocrystals are densely packed into pellets, direct interfacial connections wetted by water at the atomic level facilitate Zn²⁺ ion transport between crystals. Finally, we believe that the perfect size match between the Zn(H₂O)₆²⁺ ion (5.1 Å) and the pore of the framework (7.0 Å) plays a crucial role in ensuring excellent ion conductivity. Given its high conductivity and low activation energy, we assert that **1**^{Zn} can be defined as a superionic solid-state conductor for ZIBs.

We subsequently assessed the compatibility of **1**^{Zn} with the metallic Zn anode via continuous plating and stripping cycles. Initially, we constructed a Zn/**1**^{Zn}/Cu asymmetric cell to examine the Coulombic efficiency of Zn plating and stripping on Cu foils. As shown in Figure S23, the asymmetric Zn/**1**^{Zn}/Cu cell produced a high efficiency of 99% and could maintain stability over 200 cycles. Furthermore, during this long-term cycling experiment, the Zn/**1**^{Zn}/Cu cell exhibited a remarkably low voltage hysteresis of just 40 mV. The nucleation overpotential (NOP) of this asymmetric cell was also evaluated and found to be as low as 50 mV (Figure S24). We further assembled a symmetric Zn/**1**^{Zn}/Zn cell to evaluate its electrochemical performance and the long-term cycling stability between the Zn metal anode and **1**^{Zn} electrolyte. CV measurement revealed that the initial deposition potential of Zn²⁺ is only 42 mV (Figure S25), thereby facilitating Zn²⁺ ion transport within the anionic channel and subsequent Zn²⁺ ion deposition onto zinc foils with a lower nucleation barrier. Chronoamperometry (CA) measurement was then performed at an overpotential of −150 mV for a duration of 600 s to study the influence of **1**^{Zn} on Zn nucleation/deposition mechanisms. As shown in Figure S26, the current density exhibited a rapid increase during the initial 10 s and subsequently stabilized throughout the entire plating process, conforming to a uniform nucleation/deposition process without Zn dendrite formation on the surface of the zinc electrode. Lastly, a galvanostatic cycling (GC) measurement was conducted to assess the long-term durability at a high current of 10 mA cm^{−2} with a plating capacity of 5 mAh cm^{−2} at room temperature. As shown in Figure 2d, the GC curve of the symmetric cell produced an overpotential of less than 50 mV and maintained stable performance for over 450 h without significant fluctuations in polarization potential. The enlarged voltage profile showed a flat voltage plateau in the hole charge–discharge cycles, reflecting a uniform Zn-ion flux directed by **1**^{Zn}. Encouraged by the low overpotential, we then explored the rate performances with a fixed deposition capacity of 1 mAh cm^{−2} at various current densities from 20 to 50 mA cm^{−2}. As shown in Figure 2e, even at a super high current density of 50 mA cm^{−2}, a stable cell voltage of around 0.25 V was obtained without causing a short circuit, and the cell voltage could quickly return to its original value of less than 0.1 V upon reverting the current density back to 20 mA cm^{−2}.

We further tracked the resistance of the symmetrical cell to assess calendar aging. The total resistance (6.77 ± 0.03 Ω)

remained nearly unchanged throughout the process (Figure S29). Finally, we disassembled the symmetric cell to investigate the composition and crystallinity integrity of the **1**^{Zn} electrolyte and the surface morphology change of the Zn electrode. PXRD analysis showed that the recovered **1**^{Zn} electrolyte still retained a high degree of crystallinity after the cycling tests (Figure S8). ICP-MS measurement indicated that the Zn²⁺ content remained nearly consistent with its pristine counterpart. N₂ sorption measurements revealed that the BET surface area of the recovery was estimated to be 561 m² g^{−1} (Figure S11). SEM analysis of the postcycling Zn electrode still displayed a clean and flat surface morphology (Figure S30), visually indicating that single Zn²⁺ ion migration within the anionic framework can avoid the polarization effect and inhibit zinc dendrite formation.

To evaluate the feasibility of **1**^{Zn} in a practical battery system, we fabricated a CR2025-type coin cell utilizing **1**^{Zn}, zinc foil, and NH₄V₄O₁₀ as the electrolyte, anode, and cathode, respectively (Figure 3a). In this cell, we controlled the thickness of the **1**^{Zn} electrolyte to 1–2 mm, and the **1**^{Zn} electrolyte was directly compacted with NH₄V₄O₁₀ and polished Zn metal in an open-air environment without any special treatment for all three components. It should be noted that the fabrication of our solid ZIB is facile and efficient, and it does not need to be performed under a water- and/or oxygen-free environment or any other protection measures, making it particularly ideal for large-scale fabrication.

Figure 3g presents the initial five CV curves of our solid ZIB recorded at 1 mV s^{−1} within the voltage range of 0.2–1.8 V, in which three pairs of reduction/oxidation peaks can be observed, indicating the multistep reversible insertion/extraction process occurring in the NH₄V₄O₁₀ cathode. Among these redox peaks, two distinct couples at 0.55/0.64 V and 0.92/1.14 V corresponded to the redox reactions of V⁴⁺/V³⁺ and V⁵⁺/V⁴⁺,⁴⁷ respectively, signifying the multistep Zn²⁺ intercalation/deintercalation process. Additionally, a pair of small redox peaks at 1.32/1.27 V might be related to the oxidation–reduction process of NH₄⁺ in V–O interlayers,^{48–52} while a gradually disappeared oxidative peak at 1.41 V, likely ascribed to the complete phase transition of NH₄V₄O₁₀.⁵² The galvanostatic charge–discharge (GCD) performance under varied current densities at room temperature is illustrated in Figure 3c,d, in which our solid ZIB released a high reversible capacity of 497 mAh g^{−1} at a current density of 0.3 A g^{−1} and could retain the capacities of 437, 394, 351, and 274 mAh g^{−1} at the current densities of 1, 3, 5, and 10 A g^{−1}, respectively. Notably, even at a super high rate of 100 A g^{−1}, our ZIB could still deliver a capacity of 47 mAh g^{−1} and rapidly recover back to 490 mAh g^{−1} when the current density was cycled back to 0.3 A g^{−1} (Figure 3c). We further examined its rate performance across a broad temperature range. Figure 3e,3f displays its rate capability from −40 to 60 °C. At 60 °C, the battery delivered capacities of 460, 406, 370, and 288 mAh g^{−1} at current densities of 1, 2, 5, and 10 A g^{−1} (Figures 3f and S31), respectively. When the temperature was decreased to −20 °C, the battery still exhibited capacities of 213 and 51 mAh g^{−1} at 1 and 10 A g^{−1}, respectively. Upon returning to 60 °C, the battery showed a significant capacity recovery, reaching 450 mAh g^{−1} (Figure S32).

The long-term GCD stability was subsequently assessed at the current density of 5 A g^{−1} and displayed in Figure 3h. Our solid ZIB exhibited a capacity of 349 mAh g^{−1} and demonstrated an admirable capacity retention of 99.7% even

after 2500 cycles, accompanied by a Coulombic efficiency approaching 100% and tiny resistance rises (Figure S33). Furthermore, even at high current densities of 50, 70, and 100 A g⁻¹, our solid ZIB still delivered high Coulombic efficiency and capacity retention, both approximately reaching 100% after the same number of cycles (Figure S34). We also evaluated the long-term GCD performances at a current density of 5 A g⁻¹ at -40 and 60 °C, respectively. As shown in Figure S36, our solid ZIB could maintain capacities of 126 and 436 mAh g⁻¹, respectively. Even after 2200 cycles, it could still retain 93 and 96% of its initial capacities, respectively, with both Coulombic efficiencies achieving nearly 100%. More encouragingly, at some other more extreme conditions—such as 10 A g⁻¹ at -20 °C and 20 A g⁻¹ at -10 °C—our solid ZIBs still exhibited exceptional resilience and were capable of undergoing more than 1000 cycles with 97.3 and 94.3% capacity retention versus their initial capacity, respectively (Figures S37 and S38), further underscoring their superlong lifespan with extremely high stability.

The self-discharge behavior of our solid ZIB was then studied by precharging to 1.6 V first. As shown in Figure S39, our solid ZIB maintained a stable voltage of 1.33 V after resting for 24 h at room temperature and delivered a discharge capacity of 388 mAh g⁻¹ with a 93% capacity retention. Given its excellent charge and discharge performance, the brilliant red-light-emitting diodes (LEDs) could be briefly powered using our solid ZIB. After cycling and disassembling the solid ZIB, the anode zinc sheet remained as smooth and bright as before, and the MOF solid electrolyte also stayed intact, with no cracks or voids being detected (Figure 3h, insets). Finally, we assembled a pouch cell to verify its practical application. As shown in Figure S40, the pouch cell outputs an open-circuit potential of 1.14 V. Three pouch cells could power more than 77 LEDs (Figure S41), with no observable abnormality even at temperatures of 60 °C and -20 °C (Figure S42).

Our solid electrolyte 1^{Zn} could further tolerate other cathode materials, such as MnO₂, PTCDA, and Cu₃(HHTP)₂, enabling the formation of stable solid ZIBs that achieved capacities of 65, 190, and 70 mAh g⁻¹, respectively. These three solid ZIBs were capable of undergoing more than 1000 consecutive charge/discharge cycles at room temperature (Figures S43, S49, and S55). Therefore, we concluded that our anionic MOF electrode not only facilitated efficient Zn²⁺ transport in its anionic channels but also exhibited excellent interfacial compatibility with Zn anode and cathode materials like NH₄V₄O₁₀, MnO₂, PTCDA, and Cu₃(HHTP)₂, enabling effective utilization of Zn metal and stable rate performance during the continuous plating/stripping and charge/discharge cycles.

CONCLUSIONS

In summary, by employing the carboxylic acid functionalized anionic tetraphenylborate ligand, we have constructed a highly stable anionic metal–organic framework. After the Zn²⁺ ion exchange of the leaving freedom Li⁺ in the anionic pore, the resulting anionic MOF delivers outstanding Zn²⁺ conductivity, impressively high transference number, low activation energy, good interfacial compatibility, and excellent rate performance. The regular and well-ordered pores, combined with the uniform distribution of BPh₄⁻ anions throughout the crystals, fundamentally facilitate Zn²⁺ rapid transport and deposition. All these properties consequently enable the high performance of the following solid-state zinc-ion battery with good rate

performance and long-term cycling stability, thereby making our battery have great potential for commercial applications. Moreover, our anionic metal–organic framework can also tolerate other ions such as Na⁺, K⁺, Mg²⁺, Ca²⁺, or Al³⁺, which may make it suitable for other new types of solid-state batteries in the future.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/jacs.Sc08566>.

Details on experimental procedures, X-ray crystallographic data for 1^{Zn}, supporting figures, reaction procedures, and spectra (PDF)

Accession Codes

Deposition Number 2404118 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge via the joint Cambridge Crystallographic Data Centre (CCDC) and Fachinformationszentrum Karlsruhe Access Structures service.

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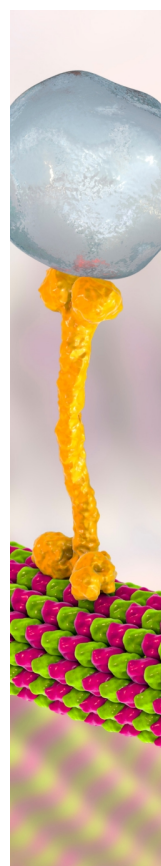
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